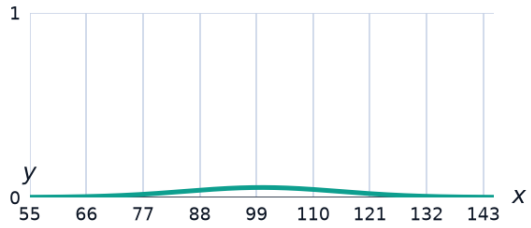
Normal Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 19.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

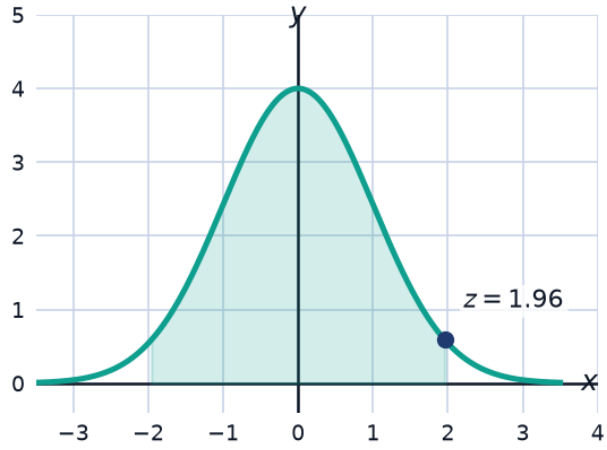
1 **[Foundational] • Target: Finding probabilities using the standard normal table**

A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.



2 **[Foundational] • Target: Properties of the normal distribution**

State two key properties of the normal distribution curve.



3 **[Foundational] • Target: Properties of the normal distribution**

Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.

4 **[Foundational] • Target: Properties of the normal distribution**

State one similarity and one difference between the normal distribution and the discrete uniform distribution.

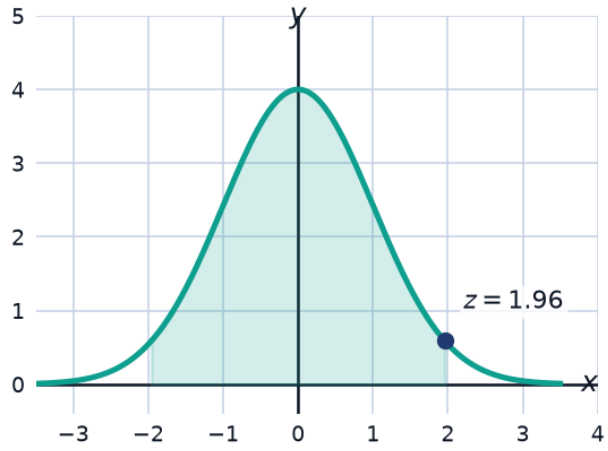
5 **[Foundational] • Target: Standardizing a normal variable**

A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x - \mu}{\sigma}$.

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6 [Foundational] • Target: Standardizing a normal variable

A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.



7 [Foundational] • Target: Standardizing a normal variable

A random variable $X \sim N(70, 144)$. Find the z-score corresponding to $x = 88$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding probabilities using the standard normal table

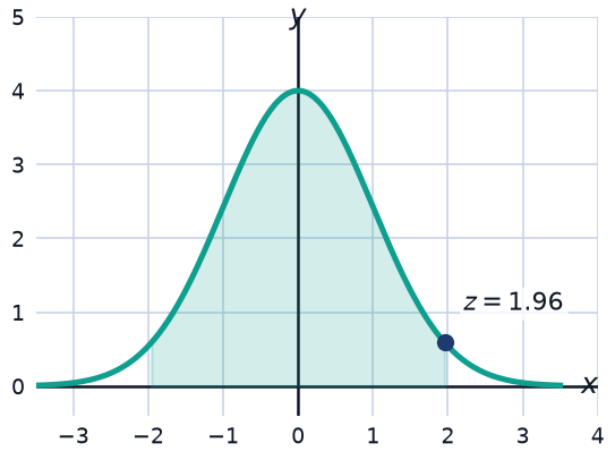
Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.

9 [Intermediate] • Target: Finding probabilities using the standard normal table

A random variable $X \sim N(50, 16)$. Find $P(X < 54)$, given that $\Phi(1) \approx 0.8413$.

10 [Intermediate] • Target: Finding probabilities using the standard normal table

A random variable $X \sim N(80, 100)$. Find $P(X > 95)$, given that $\Phi(1.5) \approx 0.9332$.



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11 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

12 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(80, 64)$. Find the value x such that $P(X < x) = 0.9772$, given that $\Phi(2) \approx 0.9772$.

13 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(45, 49)$. Find the value x such that $P(X < x) = 0.8413$, given that $\Phi(1) \approx 0.8413$.

14 [Intermediate] • Target: The probability of a value lying between two bounds

A random variable $X \sim N(40, 25)$. Find $P(35 < X < 48)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.6) \approx 0.9452$.

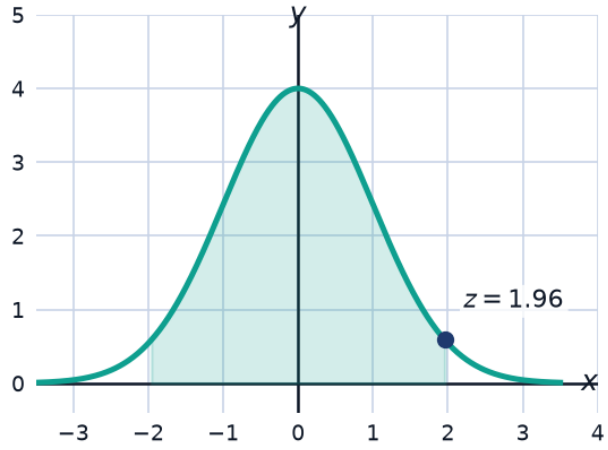
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The probability of a value lying between two bounds • Cross-Lesson: 15.01 Representation of Data

A random variable $X \sim N(20, 16)$. Find $P(16 < X < 26)$, given that $\Phi(1) \approx 0.8413$ and $\Phi(1.5) \approx 0.9332$.

16 [Challenging] • Target: The probability of a value lying between two bounds • Cross-Lesson: 16.01 Permutations & Combinations

A random variable $X \sim N(50, 100)$. Find $P(35 < X < 65)$, given that $\Phi(1.5) \approx 0.9332$.



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17 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 17.01 Discrete Random Variables

A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1-p)$ should exceed 5).

18 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 18.01 Binomial Distribution

A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

19 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A random variable $X \sim B(100, 0.3)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

20 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 3.01 Graphs & Transformations

A random variable $X \sim B(250, 0.2)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(Z < 1.645)$ for the standard normal distribution $Z \sim N(0, 1)$.

a. 0.9500

b. 0.9750

c. 0.9000

d. 0.9900

22 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(Z > -1.282)$ for $Z \sim N(0, 1)$.

a. 0.9000

b. 0.1000

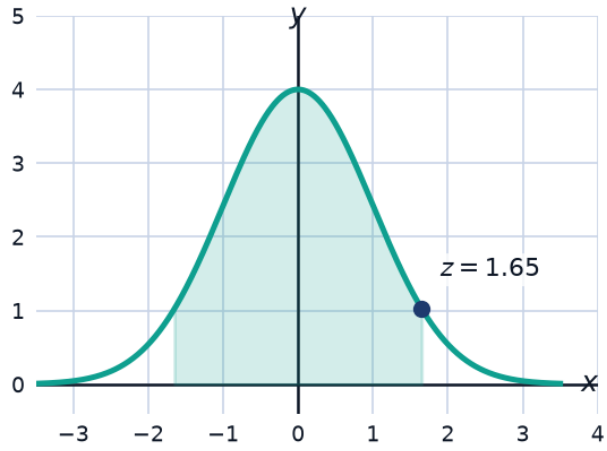
c. 0.8000

d. 0.8500

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23 [Foundational] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$

Find $P(-1.96 < Z < 1.96)$ for $Z \sim N(0, 1)$.



a. 0.9500

b. 0.9750

c. 0.9000

d. 0.9900

24 [Foundational] • Target: Standardization & General Normal Distribution

The random variable $X \sim N(50, 16)$. Find $P(X < 56)$.

a. 0.9332

b. 0.8413

c. 0.9772

d. 0.8849

25 [Foundational] • Target: Standardization & General Normal Distribution

Heights of adult females are normally distributed with mean 165 cm and standard deviation 6 cm. What percentage of females are taller than 177 cm?

a. 2.28%

b. 5.00%

c. 4.56%

d. 1.14%

26 [Foundational] • Target: Standardization & General Normal Distribution

If $Y \sim N(100, 225)$, find $P(85 < Y < 115)$.

a. 0.6826

b. 0.9544

c. 0.5000

d. 0.7500

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27 [Foundational] • Target: Inverse Normal & Finding μ and σ

The random variable $X \sim N(\mu, 25)$ has $P(X < 62) = 0.8413$. Find the mean μ.

a. 57

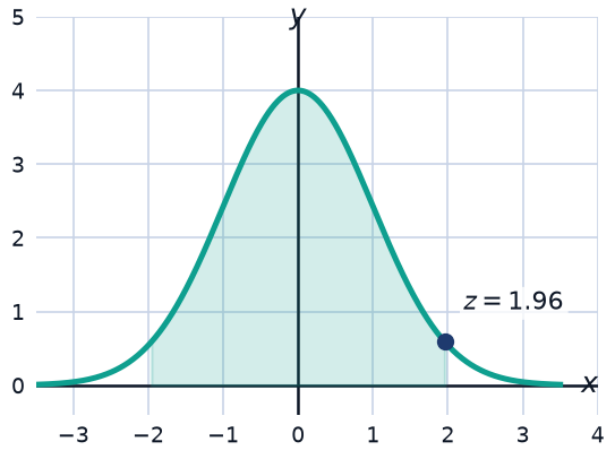
b. 55

c. 52

d. 60

28 [Foundational] • Target: Inverse Normal & Finding μ and σ

For a normal variable X, $P(X < 40) = 0.05$ and $P(X < 70) = 0.95$. Find μ and σ.



a. $\mu = 55, \sigma = 9.12$

b. $\mu = 55, \sigma = 15.0$

c. $\mu = 50, \sigma = 10.0$

d. $\mu = 55, \sigma = 5.50$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inverse Normal & Finding μ and σ

Packets of biscuits have nominal weight 200 g. Actual weights are normally distributed with $\mu = 204$ g and $\sigma = 2.5$ g. What proportion of packets are underweight (< 200 g)?

a. 0.0548

b. 0.0228

c. 0.0100

d. 0.0808

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30 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

State the two conditions required to approximate a binomial distribution $X \sim B(n, p)$ by a normal distribution.

a. $np > 5$ and $n(1 - p) > 5$

b. $n > 30$ and $p = 0.5$

c. $np > 10$ only

d. $n(1 - p) > 100$

31 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

A fair coin is tossed 100 times. Using the normal approximation with continuity correction, what is the expression for $P(X \geq 55)$?

a. $P(Y > 54.5)$ where $Y \sim N(50, 25)$

b. $P(Y > 55.5)$ where $Y \sim N(50, 25)$

c. $P(Y > 54.5)$ where $Y \sim N(50, 5)$

d. $P(Y > 55.0)$ where $Y \sim N(50, 25)$

32 [Intermediate] • Target: Normal Approximation to the Binomial Distribution

Compute the approximate probability of getting at least 55 heads in 100 tosses using $Y \sim N(50, 25)$.

a. 0.1841

b. 0.1587

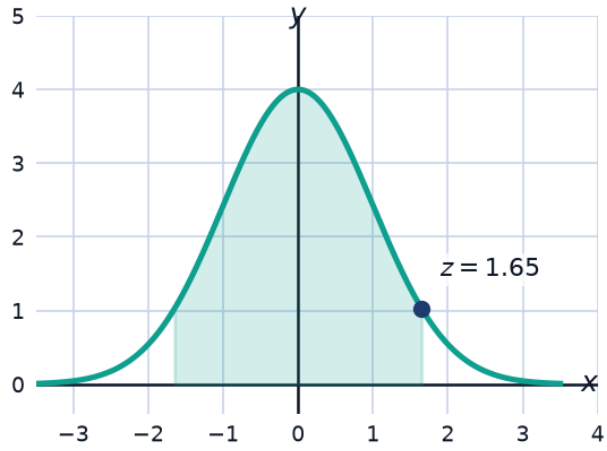
c. 0.2119

d. 0.1250

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33 [Intermediate] • Target: Properties of the normal distribution

Given $X \sim N(\mu, \sigma^2)$, $P(X < 57) = 0.9$ and $P(X < 27) = 0.1$, find μ, correct to the nearest integer.



a. 39

b. 42

c. 48

d. 45

34 [Intermediate] • Target: Standardizing a normal variable

Given $X \sim N(\mu, \sigma^2)$, $P(X < 100) = 0.95$ and $P(X < 80) = 0.05$, find μ, correct to the nearest integer.

a. 87

b. 93

c. 96

d. 90

35 [Intermediate] • Target: Finding probabilities using the standard normal table

Given $X \sim N(\mu, \sigma^2)$, $P(X < 51) = 0.8$ and $P(X < 37) = 0.2$, find μ, correct to the nearest integer.

a. 50

b. 41

c. 47

d. 44

36 [Intermediate] • Target: Finding a value given a probability (inverse normal)

Given $X \sim N(\mu, \sigma^2)$, $P(X < 46) = 0.8$ and $P(X < 32) = 0.2$, find μ, correct to the nearest integer.

a. 45

b. 39

c. 42

d. 36

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37 [Intermediate] • Target: The probability of a value lying between two bounds

Given $X \sim N(\mu, \sigma^2)$, $P(X < 107) = 0.9$ and $P(X < 77) = 0.1$, find μ, correct to the nearest integer.

a. 92

b. 98

c. 89

d. 95

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 18.01 Binomial Distribution

Given $X \sim N(\mu, \sigma^2)$, $P(X < 55) = 0.9$ and $P(X < 29) = 0.1$, find μ, correct to the nearest integer.

a. 45

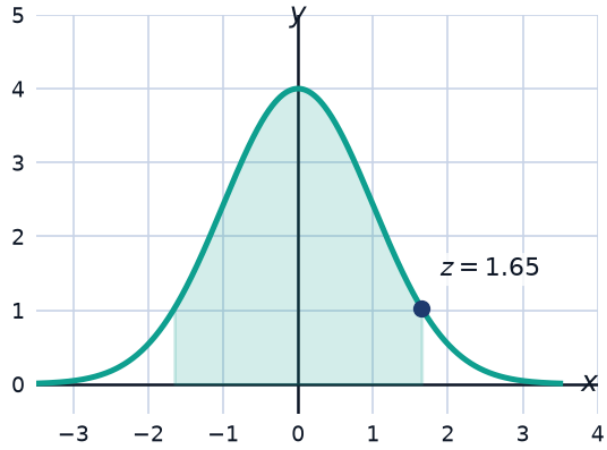
b. 42

c. 39

d. 48

39 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Given $X \sim N(\mu, \sigma^2)$, $P(X < 36) = 0.85$ and $P(X < 12) = 0.15$, find μ, correct to the nearest integer.



a. 27

b. 24

c. 30

d. 21

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40 [Challenging] • Target: A well-designed word problem: setting a pass mark for an exam • Cross-Lesson: 3.01 Graphs & Transformations

Given $X \sim N(\mu, \sigma^2)$, $P(X < 103) = 0.9$ and $P(X < 77) = 0.1$, find μ, correct to the nearest integer.

a. 96

b. 87

c. 93

d. 90

41 [Challenging] • Target: The Standard Normal Distribution $Z \sim N(0, 1)$ • Cross-Lesson: 4.01 Coordinate Geometry

Given $X \sim N(\mu, \sigma^2)$, $P(X < 99) = 0.8$ and $P(X < 83) = 0.2$, find μ, correct to the nearest integer.

a. 94

b. 91

c. 88

d. 97

42 [Challenging] • Target: Standardization & General Normal Distribution • Cross-Lesson: 5.01 Trigonometry

Given $X \sim N(\mu, \sigma^2)$, $P(X < 66) = 0.9$ and $P(X < 36) = 0.1$, find μ, correct to the nearest integer.

a. 48

b. 57

c. 54

d. 51

43 [Challenging] • Target: Inverse Normal & Finding μ and σ • Cross-Lesson: 6.01 Differentiation & Integration

A random variable $X \sim B(201, 0.75)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 148.8

b. 152.8

c. 155.8

d. 150.75

44 [Challenging] • Target: Normal Approximation to the Binomial Distribution • Cross-Lesson: 7.01 Statistics & Probability

A random variable $X \sim B(266, 0.7)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 184.2

b. 186.2

c. 188.2

d. 191.2

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45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 8.01 Functions: Composite & Inverse

A random variable $X \sim B(170, 0.25)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 44.5

b. 47.5

c. 42.5

d. 40.5

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